Manip — cheat sheet

Keyboard-driven algebraic manipulation. Lowercase letters, digits and symbols apply operations directly to the formula (the same keys work in the Enter menu); uppercase (Shift) are commands.

NAVIGATION AND STEPS

← → ↑ ↓	navigate
Ctrl ← →	move / take to the other side
Ctrl ↑ ↓	leave / enter group
Shift ← →	select neighbors
Space	select
Esc	clear selection / leave
Enter	operations
Tab	workspaces
PgUp PgDn Home End	go through the steps
Ctrl Z	undo
Ctrl Shift Z	redo

COMMANDS (SHIFT)

C	color
E	edit
L	copy LaTeX
P	present
K	keyboard
O	step note
R	record video
]	larger formula
[	smaller formula
\	normal size
?	shortcuts panel

ARITHMETIC

c	Calculate	$x = 7 - 3 \rightarrow x = 4$
l	Combine like terms	$x = 2a + 3a \rightarrow x = 5a$
x	Cancel	$x + 3 - 3 = 5 \rightarrow x = 5$
0	Remove identity element	$x + 0 = y \rightarrow x = y$
r	Sign rule	$x = a - (-b) \rightarrow x = a + b$

STRUCTURE

d	Distribute	$2(x+1)=4 \rightarrow 2x+2\cdot 1=4$
e	Factor out	$2x+2y=0 \rightarrow 2(x+y)=0$
(	Wrap in parentheses	$a+b+c=0 \rightarrow a+(b+c)=0$ with selection
p	Remove parentheses	$a+(b+c)=0 \rightarrow a+b+c=0$
g	Group selection	$a+b+c=0 \rightarrow a+(b+c)=0$ with selection
q	Add and subtract...	$x^2+6x=7 \rightarrow x^2+6x+9-9=7$ typing “9”
s	Substitute with...	$x+y=10 \rightarrow x+2x=10$ typing “2x”

FRACTIONS

j	Combine fractions	$\frac{x}{2}+\frac{y}{2}=1 \rightarrow \frac{x+y}{2}=1$
f	Split fraction	$\frac{x+y}{2}=1 \rightarrow \frac{x}{2}+\frac{y}{2}=1$

POWERS

o	Power rule	$x^2x^3=1 \rightarrow x^{2+3}=1$
u	Write as a power	$aab=0 \rightarrow a^2b=0$
m	Write power as a product	$(a+b)^2=0 \rightarrow (a+b)(a+b)=0$
b	Expand power (binomial / multinomial)	$(a+b)^3=0 \rightarrow a^3+3a^2b+3ab^2+b^3=0$
2	Special product: $a^2 \pm 2ab + b^2$, $a^2 - b^2$	$x^2+6x+9=0 \rightarrow (x+3)^2=0$

COMPLEX NUMBERS

z	Conjugate properties	$w=\overline{ab} \rightarrow w=\overline{a}\,\overline{b}$
~	z ↔ $\bar{\bar{z}}$ (double conjugate)	$w=\bar{\bar{z}} \rightarrow w=z$
h	z·$\bar{z}$ ↔ $\lvert z\rvert^2$	$w=z\bar{z} \rightarrow w=\lvert z\rvert^2$
v	z ± $\bar{z}$ ↔ 2Re(z) / 2i·Im(z)	$w=z+\bar{z} \rightarrow w=2\,\mathrm{Re}(z)$
y	Multiply by the conjugate	$w=\frac{1}{3+4\mathrm{i}} \rightarrow w=\frac{1(3-4\mathrm{i})}{(3+4\mathrm{i})(3-4\mathrm{i})}$

EQUATION AND INEQUALITY

=	Apply to both sides...	$2x=4 \rightarrow \frac{2x}{2}=\frac{4}{2}$ typing “/ 2”
t	Swap sides	$x<4 \rightarrow 4>x$
i	Estimate (≤, ≥, <, >)...	$y=x^2+1 \rightarrow y\geq 1$ typing “>= 1”

SYSTEMS

+	Add equations (select two)	$\begin{cases}x+y=3\\x-y=1\end{cases} \rightarrow \begin{cases}x+y=3\\x-y+x+y=1+3\end{cases}$ with selection
-	Subtract equations (select two)	$\begin{cases}x+y=3\\x-y=1\end{cases} \rightarrow \begin{cases}x+y=3\\x-y-(x+y)=1-3\end{cases}$ with selection
>	Substitute this variable into the other equations	$\begin{cases}y=2x+1\\x+y=7\end{cases} \rightarrow \begin{cases}y=2x+1\\x+(2x+1)=7\end{cases}$

NAMES

n	Name it...	$2(x+1)=y \rightarrow 2\boxed{K}=y$ typing “K”
k	Expand the name	
w	Open in workspace	

Generated by Manip itself: the examples are the operations actually applied.